Ketogenesis

Biochemistry Department

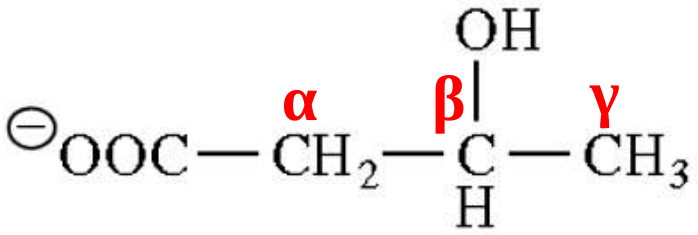
- **Increased fatty acid oxidation is a characteristic of *starvation* and of *diabetes mellitus*, leading to ketone body production by the liver (ketosis).**
- **Ketone bodies are acidic and when produced in excess over long periods, as in diabetes, cause ketoacidosis, which is ultimately fatal. Because gluconeogenesis is dependent upon fatty acid oxidation, any impairment in fatty acid oxidation leads to hypoglycemia.**

Definition

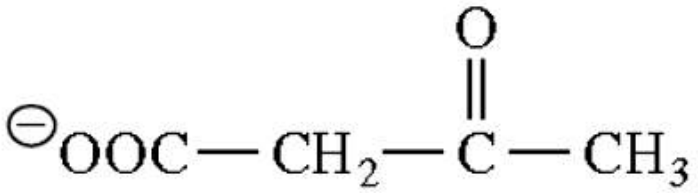
- **During high rates of fatty acid oxidation, primarily in the liver, large amounts of acetyl-CoA are generated. These exceed the capacity of the TCA cycle, and one result is the synthesis of ketone bodies, or ketogenesis.**
- **The ketone bodies are acetoacetate, β-hydroxybutyrate, and acetone.**

Ketone bodies

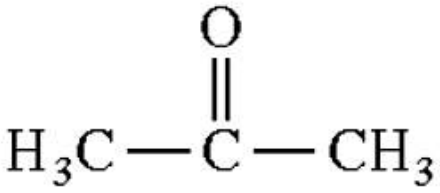
- β-hydroxybutyrate
- Acetoacetate
- acetone



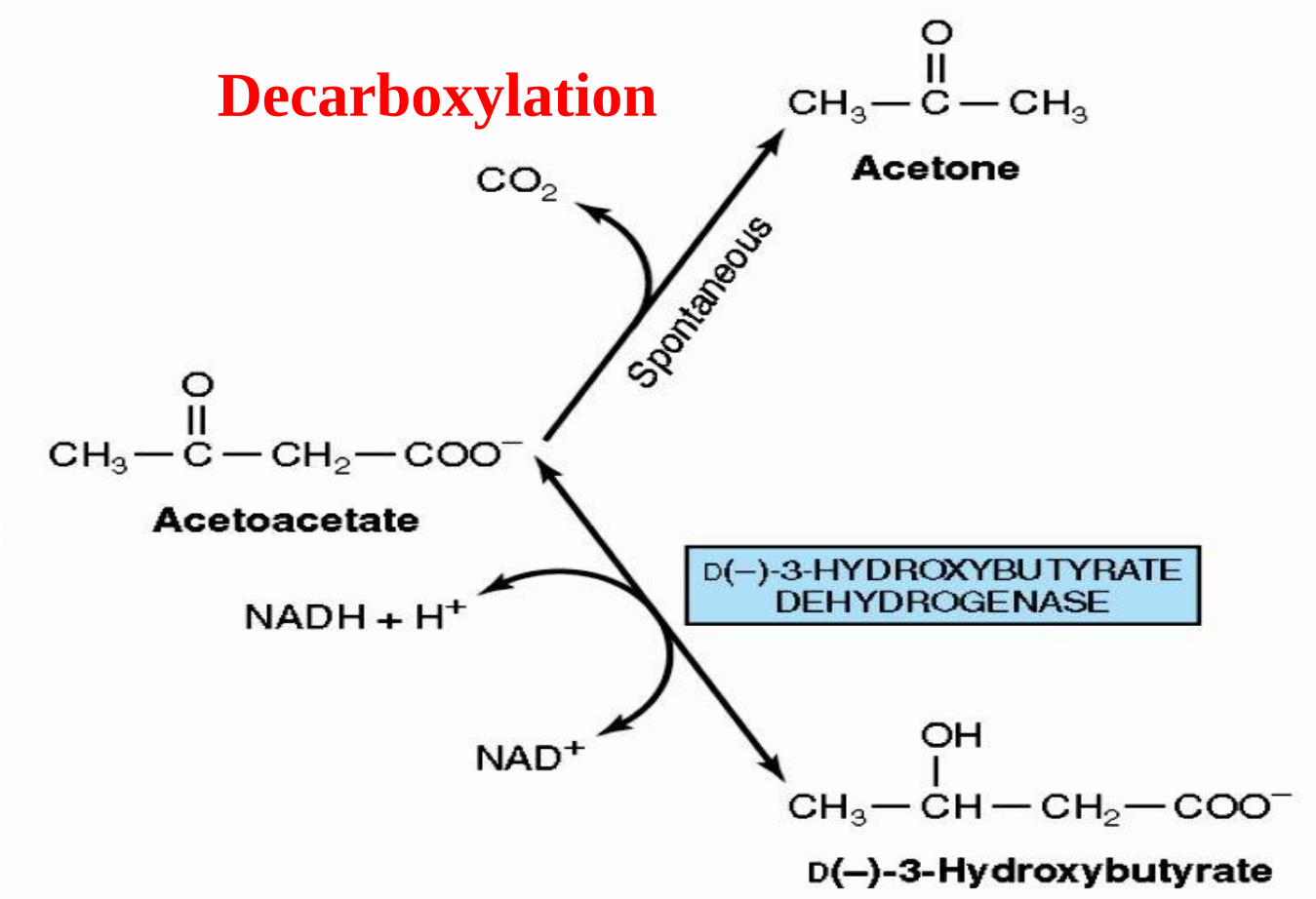
β-Hydroxybutyrate



Acetoacetate



Acetone



Interrelationships of the ketone bodies. D(−)-3-hydroxybutyrate dehydrogenase is a mitochondrial enzyme.

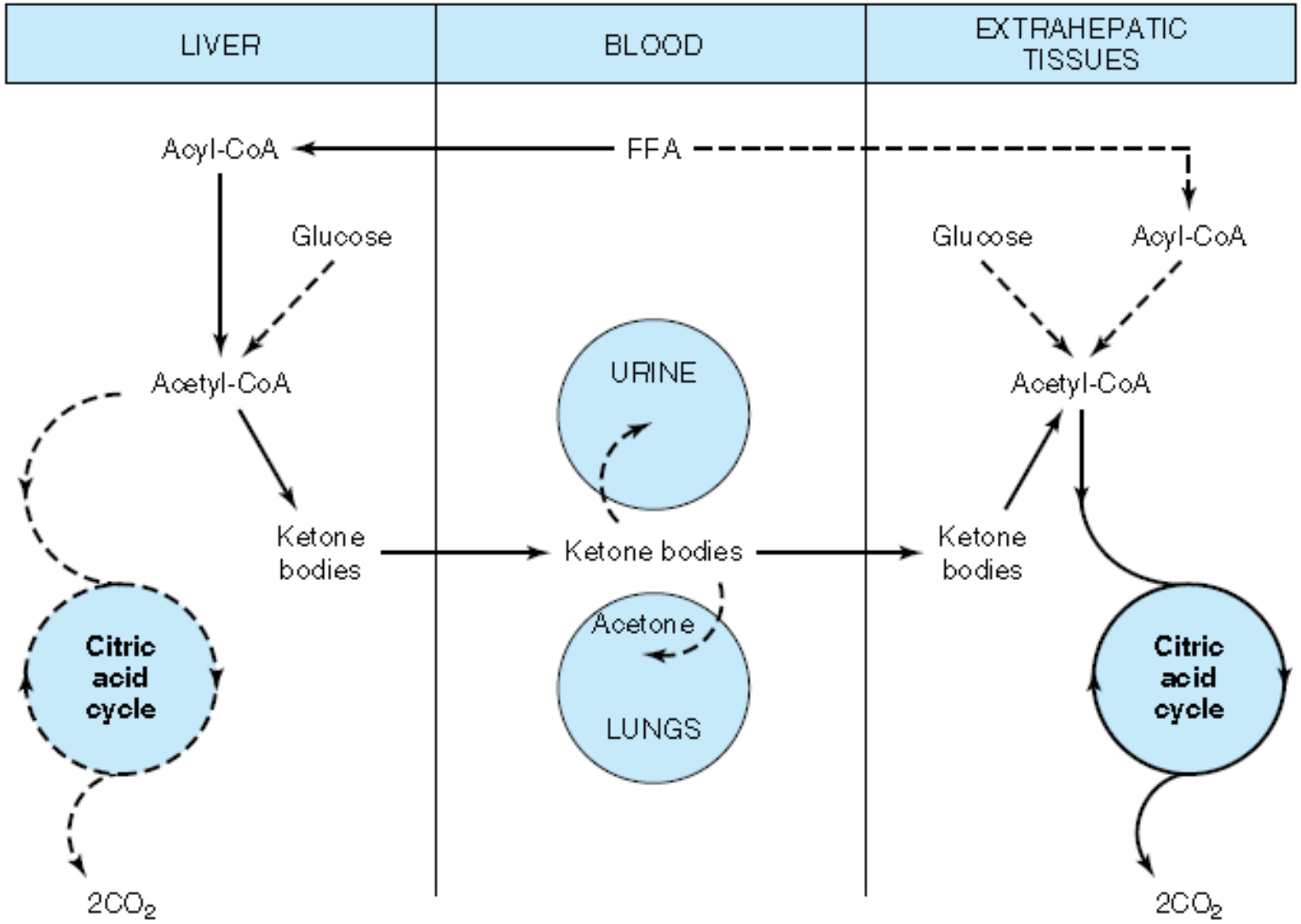
β-Hydroxybutyrate and acetoacetate are fuel molecules

- **They have less potential metabolic energy than the fatty acids from which they are derived but they make up for this deficiency by serving as "water-soluble lipids" that can be more readily transported in the blood plasma.**
- ***During starvation*, ketone bodies are produced in large amounts becoming substitutes for glucose as the principal fuel for brain cells.**
- **Ketone bodies are also metabolized in skeletal muscle and in the intestine during starvation.**

Ketone Bodies Are Synthesized in the Liver

- **In mammals, ketone bodies are synthesized in the liver and exported for use by other tissues.**
- **Ketone body synthesis :**
  - **First, two molecules of acetyl CoA condense to form acetoacetyl CoA and HS–CoA in a reaction catalyzed by acetoacetyl-CoA thiolase.**
  - **Subsequently, a third molecule of acetyl CoA is added to acetoacetyl CoA to form 3-hydroxy-3-methylglutaryl CoA (HMG CoA) in a reaction catalyzed by HMG-CoA synthase.**

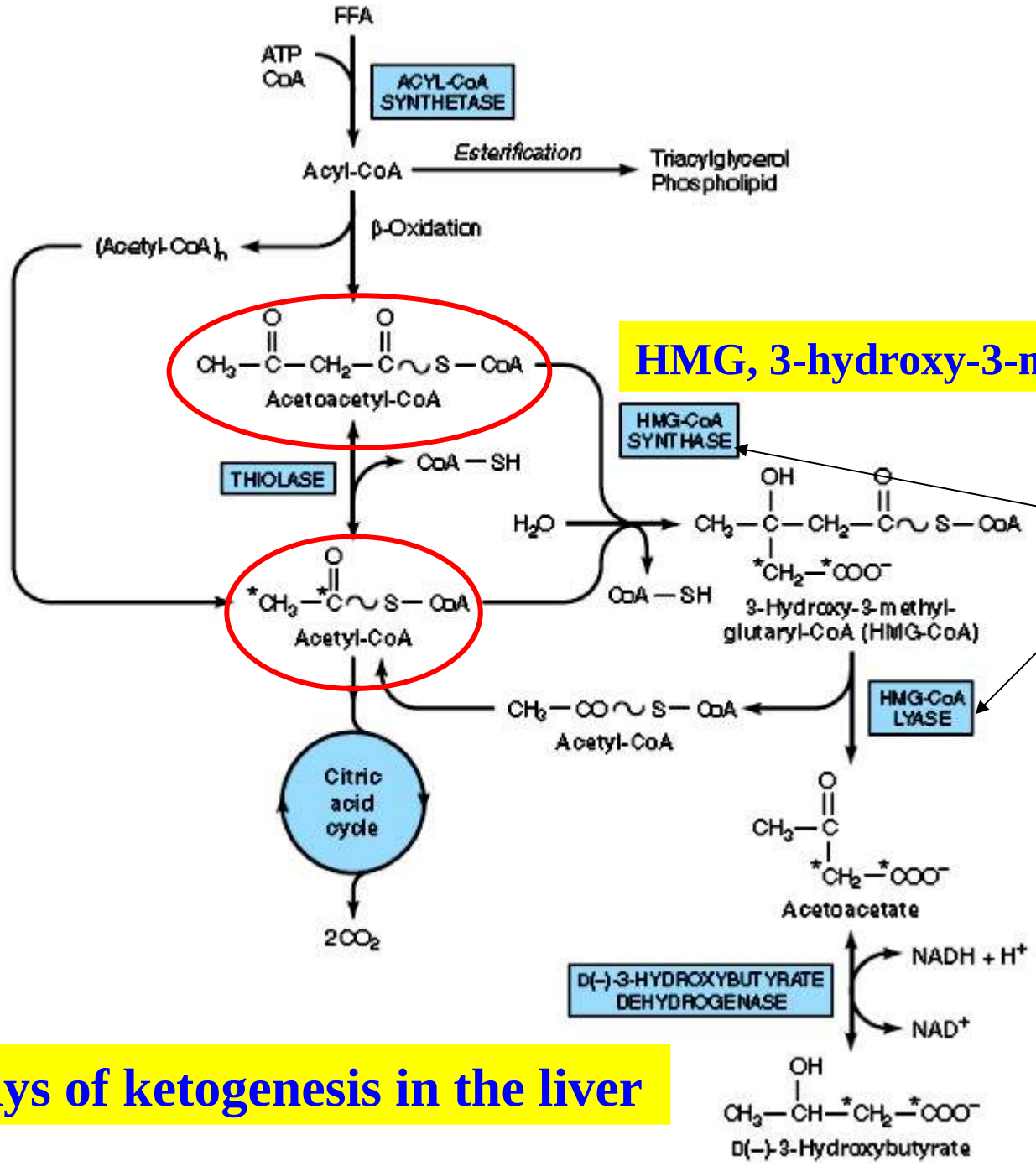
- **These steps are identical to the first two steps in the isopentenyl diphosphate biosynthesis pathway .**
- **Acetoacetate and 3-hydroxybutyrate are interconverted by the mitochondrial enzyme D(-)-3-hydroxybutyrate dehydrogenase; the equilibrium is controlled by the mitochondrial [NAD^+]/[NADH] ratio, *ie*, the redox state.**
- **Enzymes responsible for ketone body formation are associated mainly with the mitochondria.**



Formation, utilization, and excretion of ketone bodies.

- **Two acetyl-CoA molecules formed in β-oxidation condense with one another to form acetoacetyl-CoA by a reversal of the thiolase reaction.**
- **Acetoacetyl-CoA, which is the starting material for ketogenesis, also arises directly from the terminal four carbons of a fatty acid during β-oxidation.**

- **Condensation of acetoacetyl-CoA with another molecule of acetyl-CoA by 3-hydroxy-3-methylglutaryl-CoA synthase forms HMG-CoA. 3-Hydroxy-3-methylglutaryl-CoA lyase then causes acetyl-CoA to split off from the HMGCoA, leaving free acetoacetate.**



HMG, 3-hydroxy-3-methylglutaryl

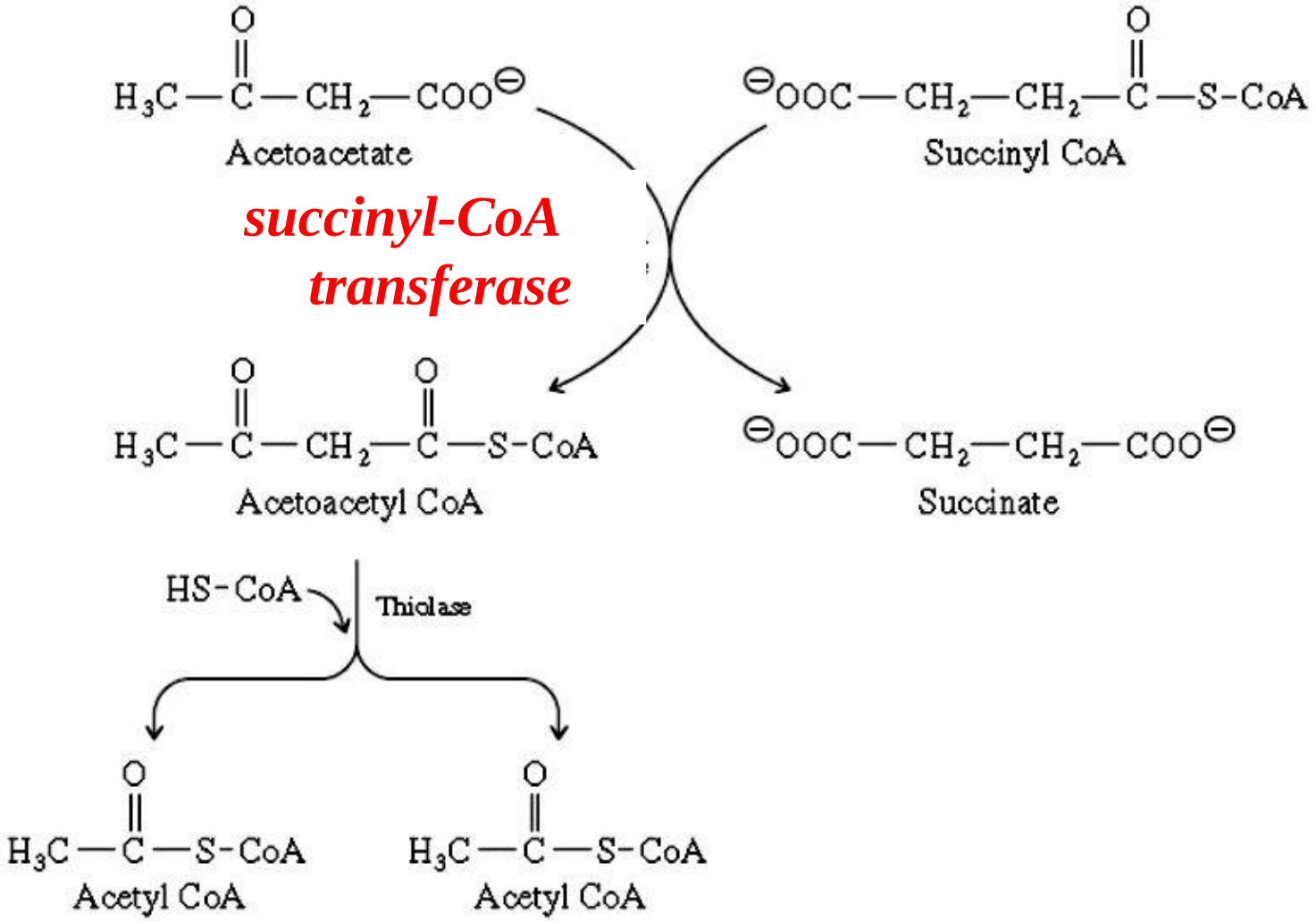
Both enzymes must be present in mitochondria for ketogenesis to take place.

Pathways of ketogenesis in the liver

Ketone Bodies Are Oxidized in Mitochondria

- **In cells that use them as an energy source, β-hydroxybutyrate and acetoacetate enter mitochondria where they are converted to acetyl CoA, which is oxidized by the citric acid cycle.**
- **β-hydroxybutyrate is converted to acetoacetate in a reaction catalyzed by an isozyme of β-hydroxybutyrate dehydrogenase that is distinct from the liver enzyme. Acetoacetate reacts with succinyl CoA to form acetoacetyl CoA in a reaction catalyzed by *succinyl-CoA transferase* .**

- **Ketone bodies are broken down only in nonhepatic tissues because this transferase is present in all tissues except live.**
- **In most cases, ketonemia is due to increased production of ketone bodies by the liver rather than to a deficiency in their utilization by extrahepatic tissues**



Conversion of acetoacetate to acetyl CoA.

Ketone Bodies Serve as a Fuel for Extrahepatic Tissues

- **In extrahepatic tissues, acetoacetate is activated to acetoacetyl-CoA by succinyl-CoA-acetoacetate CoA transferase. CoA is transferred from succinyl-CoA to form acetoacetyl-CoA. The acetoacetyl-CoA is split to acetyl-CoA by thiolase and oxidized in the citric acid cycle.**

Regulation of Ketogenesis

- **The fate of the products of fatty acid metabolism is determined by an individual's physiological status. Ketogenesis takes place primarily in the liver and may by affected by several factors:**

1. **Control in the release of free fatty acids from adipose tissue directly affects the level of ketogenesis in the liver. This is, of course, substrate-level regulation.**

2. Once fats enter the liver, they have **two distinct fates**. They may be activated to acyl-CoAs and **oxidized**, or esterified to glycerol in the **production of triacylglycerols**. If the liver has sufficient supplies of glycerol-3-phosphate, most of the fats will be turned to the production of triacylglycerols.

3. **The generation of acetyl-CoA by oxidation of fats can be completely oxidized in the TCA cycle. Therefore, if the demand for ATP is high the fate of acetyl-CoA is likely to be further oxidation to CO_2.**
4. **The level of fat oxidation is regulated hormonally through phosphorylation of ACC (*acetyl-CoA carboxylase)*, which may activate it (in response to glucagon) or inhibit it (in the case of insulin).**

Clinical Significance of Ketogenesi

- **The production of ketone bodies occurs at a relatively low rate during normal feeding and under conditions of normal physiological status. Normal physiological responses to carbohydrate shortages cause the liver to increase the production of ketone bodies from the acetyl-CoA generated from fatty acid oxidation.**
- **This allows the heart and skeletal muscles primarily to use ketone bodies for energy, thereby preserving the limited glucose for use by the brain.**

- **This physiological state, diabetic ketoacidosis (DKA), results from a reduced supply of glucose (due to a significant decline in circulating insulin) and a concomitant increase in fatty acid oxidation (due to a concomitant increase in circulating glucagon).**

Ketonemia

- **In most cases, ketonemia is due to increased production of ketone bodies by the liver rather than to a deficiency in their utilization by extrahepatic tissues. While acetoacetate and D(−)-3-hydroxybutyrate are readily oxidized by extrahepatic tissues, acetone is difficult to oxidize in vivo and to a large extent is volatilized in the lungs.**

- **In moderate ketonemia, the loss of ketone bodies via the urine is only a few percent of the total ketone body production and utilization. Since there are renal threshold-like effects (there is not a true threshold) that vary between species and individuals, measurement of the ketonemia, not the ketonuria, is the preferred method of assessing the severity of ketosis.**

Ketoacidosis Results From Prolonged Ketosis

- **Higher than normal quantities of ketone bodies present in the blood or urine constitute ketonemia (hyperketonemia) or ketonuria, respectively. The overall condition is called ketosis.**

- **Acetoacetic and 3-hydroxybutyric acids are both moderately strong acids and are buffered when present in blood or other tissues. However, their continual excretion in quantity progressively depletes the alkali reserve, causing ketoacidosis. This may be fatal in uncontrolled diabetes mellitus.**

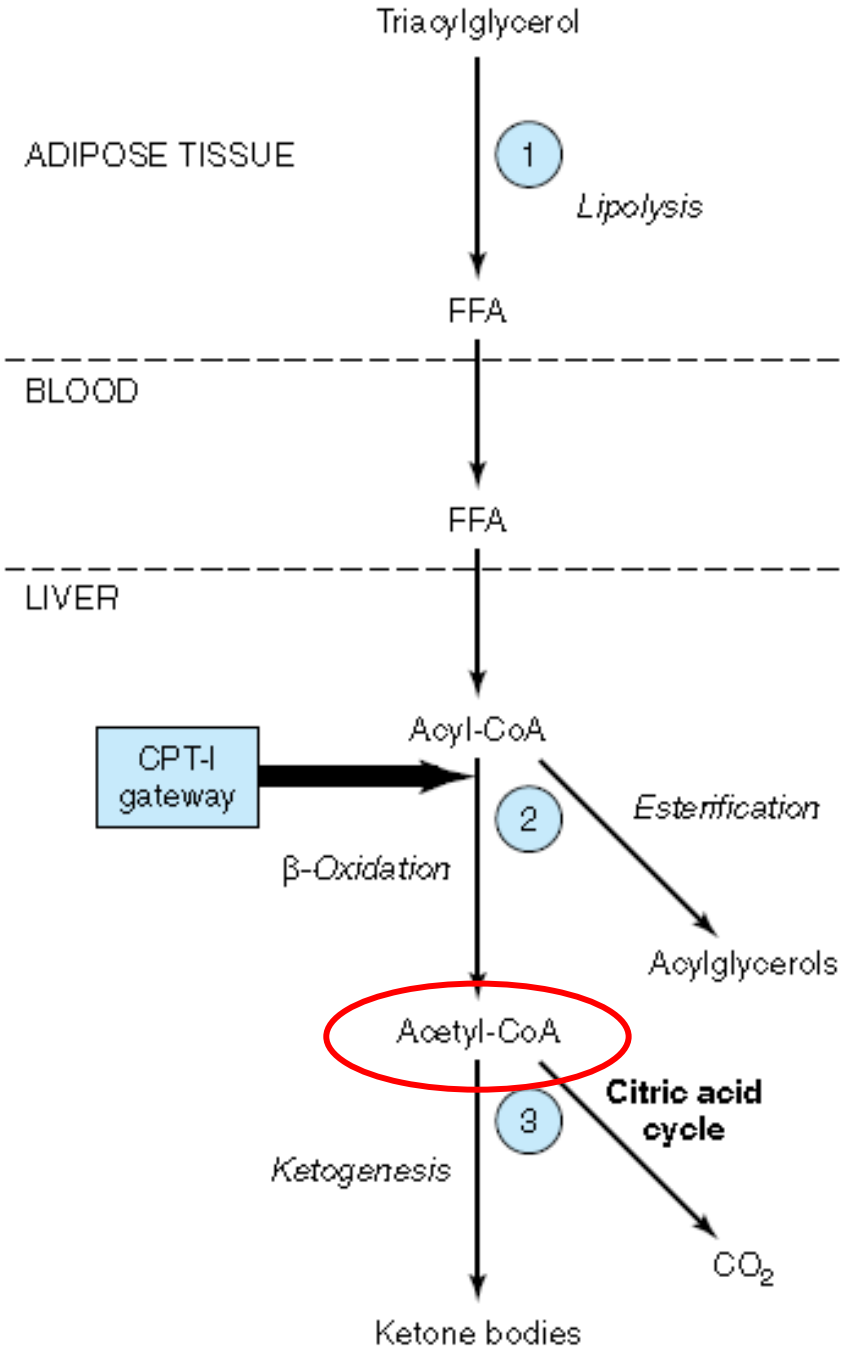
- **The basic form of ketosis occurs in starvation and involves depletion of available carbohydrate coupled with mobilization of free fatty acids. This general pattern of metabolism is exaggerated to produce the pathologic states found in diabetes mellitus.**

Ketogenesis Is Regulated At Three Crucial Steps

1. **Free fatty acids Control of free fatty acid (FFA) mobilization from adipose tissue (precursors of ketone bodies in the liver.)**
2. **The activity of carnitine acyltransferase (CAT-1) in liver, which determines the propotion of the fatty acid flux that is oxidized rather than esterified.**

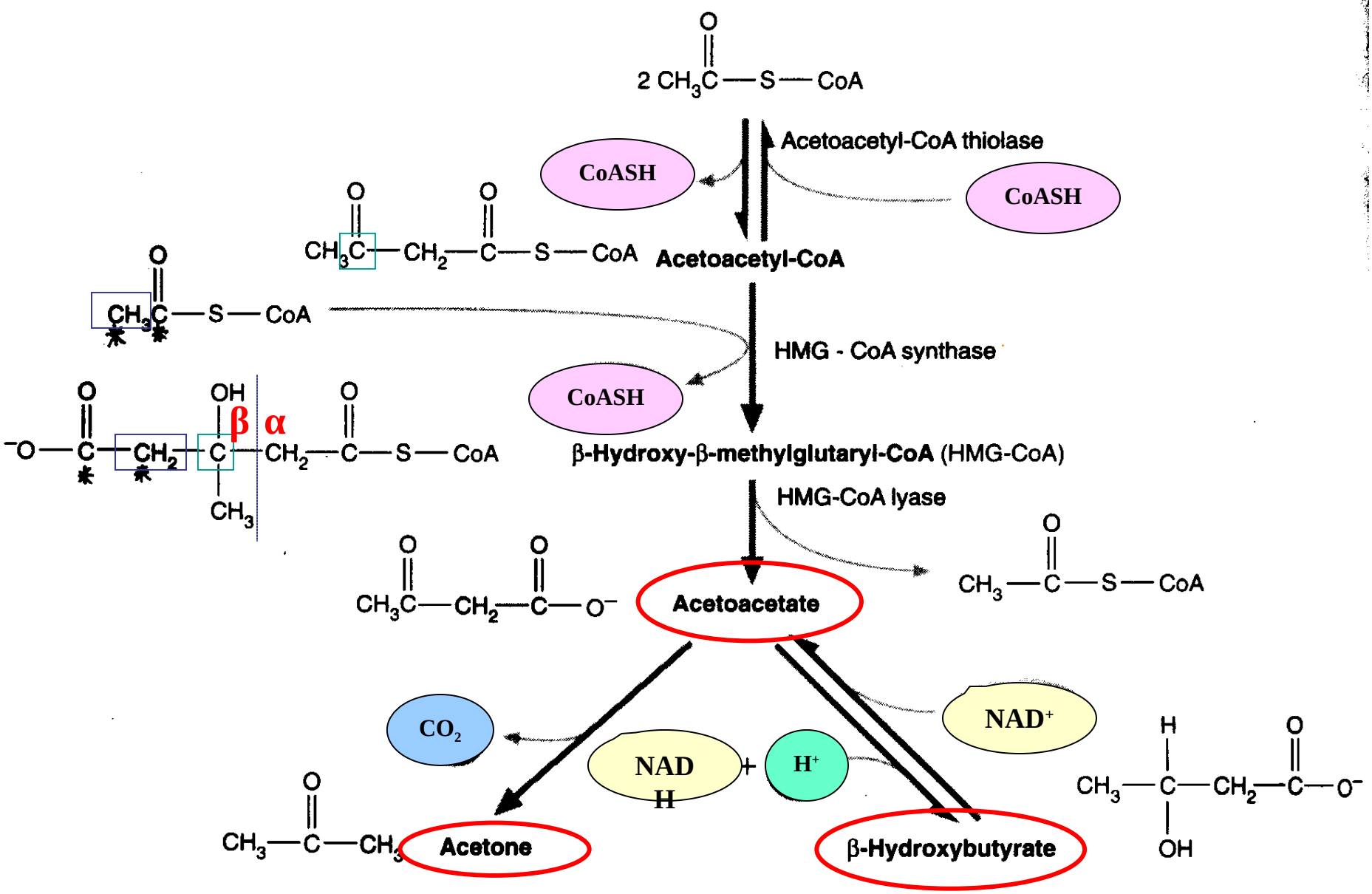
There is regulation of entry of fatty acids into the oxidative pathway by carnitine palmitoyltransferase-I (CPT-I), and the remainder of the fatty acid uptake is esterified.

3. **Partition of acetyl-CoA between the pathway of ketogenesis and the citric acid cycle**



Regulation of ketogenesis

- **A fall in concentration of oxaloacetate, particularly within the mitochondria, could impair the ability of the citric acid cycle to metabolize acetyl-CoA and divert fatty acid oxidation toward ketogenesis. Such a fall may occur because of an increase in the [NADH]/[NAD^+] ratio caused by increased β-oxidation affecting the equilibrium between oxaloacetate and malate and decreasing the concentration of oxaloacetate.**


2 CH3C—S—CoA
Acetoacetyl-CoA thiolase
CoASH
CoASH
CH3C—CH2—C—S—CoA
Acetoacetyl-CoA
CH3C—S—CoA
HMG - CoA synthase
CoASH
β α
β-Hydroxy-β-methylglutaryl-CoA (HMG-CoA)
HMG-CoA lyase
CH3C—CH2—C—O⁻
Acetoacetate
CH3—C—S—CoA
CO2
NAD+
NAD H
+
H+
CH3—C—CH3
Acetone
β-Hydroxybutyrate
CH3—C—CH2—C—O⁻

SUMMARY

1. **Fatty acid oxidation in mitochondria leads to the generation of large quantities of ATP by a process called β-oxidation that cleaves acetyl-CoA units sequentially from fatty acyl chains. The acetyl-CoA is oxidized in the citric acid cycle, generating further ATP.**

2. **The ketone bodies (acetoacetate, 3-hydroxybutyrate, and acetone) are formed in hepatic mitochondria when there is a high rate of fatty acid oxidation. The pathway of ketogenesis involves synthesis and breakdown of 3-hydroxy-3-methylglutaryl-CoA (HMGCoA) by two key enzymes, HMG-CoA synthase and HMG-CoA lyase.**

3. **Ketone bodies are important fuels in extrahepatic tissues.**
4. **Ketogenesis is regulated at three crucial steps:**
   - ① **control of free fatty acid mobilization from adipose tissue;**
   - ② **the activity of carnitine palmitoyltransferase-I in liver, which determines the proportion of the fatty acid flux that is oxidized rather than esterified;**
   - ③ **partition of acetyl-CoA between the pathway of ketogenesis and the citric acid cycle.**

5. **Ketosis is mild in starvation but severe in diabetes mellitus and ruminant ketosis.**

Thanks for your attention!